Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2425)

Key

SECTION – I

1. D
2. C
3. A
4. B
5. B
6. D
7. B
8. B
9. C
10. B
11. A
12. C
13. C
14. B
15. B
16. C
17. C
18. A
19. 4312
20. B
21. 4321
22. B
23. 4312
24. A

SECTION – II

1. D
2. A
3. D
4. C
5. D
6. B
7. 14
8. D
9. A
10. 22.5
11. 4
12. B
13. B
14. A
15. 10
16. 5
17. C
18. 4
19. D
20. C

SECTION – III

1. A
2. C
3. D
4. 16126
5. C
6. 54
7. B
8. 11
9. 6817
10. D
11. D
12. D
13. A
14. C
15. 4
16. 4
17. B
18. A
19. B
20. 25
21. B
22. 8

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 525

1. Option A: While Europe is driven by short-lived coalitions and gaseous alliances, the author does not indicate anywhere that their interests are myopic and temporary. It is more about different coalitions supporting different issues. Hence, Option A is not the answer.
Option B: The author didn't entirely discuss France and Germany losing clout. In fact, they haven't. The focus was on how they don't necessarily act as one force now, thus allowing other players to come in and influence the politics. Hence, Option B is not the answer.
Option C: The future of European politics is not temporary, leave alone something as dramatic and rhetorical as 'wisps of smoke'. It is the coalitions which are temporary in the gaseous phase. So, Option C is not the answer.
Option D: This aptly represents the point the author was trying to drive home, which is that the EU is now dominated by ever-changing coalitions instead of the permanent blocs which called the shots earlier. Hence, Option D is the answer.
Choice (D)

2. Option A: This can be understood from '***all have close links with France and Germany.***' Hence, Option A is not the answer.
Option B: This can be understood from '*all* ***can marshal coalitions of like-minded states*** *in their parts of Europe: all have modernising leaders with* ***experiences of arduous coalition-building at home.***' Hence, Option B is not the answer.
Option C: These three countries are too disparate (unequal) to make a common bloc according to the author. Hence, it cannot be inferred that they are 'desperate' to present a united front. Hence, Option C is not a similarity. Option C is the answer.
Option D: This has been mentioned partly in 'all can marshal coalitions of like-minded states in their parts of Europe'. More importantly, the author talks about the Netherlands' influence on Scandinavian and Baltic Ocean neighbors with regard to fiscal integration issues, Spain's friendships with Greece and Italy, and Austria's role as a mediator. So, it can be said that they have geopolitical clout to a certain degree. So, Option D is not the answer.
Choice (C)

© Advanced Educational Activities Pvt. Ltd. 95B, Siddamsetty Complex, Park Lane, Secunderabad – 500003.
All rights reserved. No part of this material may be reproduced, in any form or by any means, without permission in writing.
This course material is only for the use of bonafide students of Advanced Educational Activities Pvt. Ltd. (T.I.M.E.), and its licensees/franchisees and is not for sale. (11 pages) (aecg/aecf)

3. Option A: While this points to how the EU is not as powerful and dominating on the global scene as it used to be, this was not mentioned to discuss EU's changing political dynamics, which is more about internal coalitions and politics than EU's external standing. So, Option A is the answer.
Option B: This shows that Austria is becoming an important power in Europe, which was mentioned as part of the author's argument that the EU is not just dominated by two power blocks anymore; rather, it is a gaseous political alliance with shifting dynamics. Hence, Option B is not the answer.
Option C: This was mentioned by the author to prove that the Habsburg states (the Netherlands, Spain and Austria) are incapable of becoming a bloc on their own but could fluidly move and change political alliances based on issues. This was part of the author's argument that gaseous politics will define European politics based on coalitions on various issues. Hence, Option C is not the answer.
Option D: This proves that Germany and France aren't running the show together anymore, as they used to have each other's back previously. This was also part of the author's argument that European politics is shifting from the solid two-power landscape to one that is dominated by shifting alliances and coalitions on an issue-based consensus. Option D is not the answer.
Choice (A)

4. Consider the sentences: '*The Netherlands under Mark Rutte, meanwhile, has close links with both Mrs Merkel (Germany) and with Mr Macron.* ***The Netherlands has made itself influential by marshalling a "Hanseatic" alliance*** *against the fiscal integration that the likes of Mr Macron and Mr Sánchez want. In this it has allies across Scandinavian and Baltic Europe.*'
Option A: The author wanted to demonstrate how the Netherlands is gaining clout in Europe, but there is no sense of comparison with the other two countries discussed here. Hence, Option A is not the answer.
Option B: The author clearly mentions that the Netherlands has ***made itself influential*** by marshalling the alliance. So, the alliance was mentioned to show that the Netherlands is transforming into a European power that can influence policy and politics. Hence, Option B is the answer.
Option C: While the Netherlands did oppose France and Spain in the issue of fiscal integration, it shares a good rapport with both France and Germany as mentioned in the passage. So, Option C is not the answer.
Option D: Firstly, this is not factually accurate. The allies across Scandinavian and Baltic Europe are only in the one issue of fiscal integration. We are not sure about their general relationship. Secondly, even if they were all-weather allies, the author mentioned the alliance not to inform the reader about the alliance in a matter-of-fact manner, but to show that the Netherlands has influence in the region and it can drum up consensus on important issues. Hence, Option D is not the answer.
Choice (B)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 434

5. The author mentions the "value-added" model of teacher evaluations. The author makes two points in this example. The first is that how that model was based on "poor and easily fudged proxy for their overall ability". The second is that "the evaluation data is practically random."
Option A: While the author mentions that the model used was controversial, this is not the primary reason he mentioned this model. The discussion in the passage regarding this example is not about the controversial nature of this model. Hence, this is not the correct answer.
Option B: The author mentions the inadequacies of this model and how this model does not measure "harder-to-quantify" metrics. The previous paragraph also talks about poorly designed algorithms and this is an example of one. Hence, this is the primary reason for the author to mention this example.
Option C: The author does not mention that it is impossible to design algorithms for measuring performance. Hence, this is not the correct answer.
Option D: The author does not outright reject the usage of statistical models for evaluating performance. The following paragraph talks about how these models can be improved. Hence, this is not the correct answer.
Therefore, the correct answer is option B.
Choice (B)

6. The author talks about Cathy O'Neil's work from the third paragraph.
Option A: The passage mentions that "statistical models that measure performance have biases that arise from those of their creators". However, it cannot be inferred from the passage that this is intentional on part of the creator of these models. Hence, this is not a shortcoming of the models according to the passage.
Option B: O'Neil does not mention that the models ignore the data related to the metrics. She only mentions that sometimes the metrics are not chosen judiciously (as illustrated in the example provided in the penultimate paragraph). Hence, this is not the correct answer.
Option C: O'Neil does not talk about how difficult it is to create fair models. Hence, this is not the correct answer.
Option D: According to O'Neil, "statistical models that measure performance have biases that arise from those of their creators". Because of this, algorithms are "often unfair and sometimes harmful". Hence, this is the correct answer.
Choice (D)

7. The first paragraph of the passage talks about Taylor's approach for boosting productivity. The second paragraph of the passage talks about how Taylor's approach is being used.
Option A: According to Taylor, "management could ensure that their workers... did all sorts of "crude and elementary" work as efficiently as possible". However, he does not mention using statistical models for evaluating performance of workers. In the second paragraph, the author mentions that "as the metrics have become more precise in their detail, their focus has shifted beyond the tasks themselves and onto the workers doing those tasks, evaluating a broad range of their qualities". Hence, we can infer from this that Taylor's approach was extended to include performance evaluation. Therefore, this is not the correct answer.
Option B: In the first paragraph, Taylor argues that performance can be increased "by observing employees' performance and whittling down the time and effort involved in doing each task". Hence, Taylor would most probably have discussed using a data-driven approach for determining the optimum time required to complete tasks which require less skill.
Option C: The first paragraph of the passage does not mention about incentivizing the employees. The author makes a reference to "carrots and sticks" only in the second paragraph, which Taylor might not have discussed in his book.
Option D: Taylor does not talk about evaluating the performance of employees based on data-dependent approach. Hence, this is not the correct answer.
Therefore, the correct answer is option B.
Choice (B)

8. In the last para of the passage "How might these...disadvantaged backgrounds", O'Neil mentions that algorithms used to determine performance metrics are flawed and that they need to be improved, both in terms of how sound the statistical methods and also in terms of what metrics need to be taken into account while determining performance.
Looking at option A, we can clearly see that talks about using metrics such as a worker's hobbies and interests to determine what role would best suit them but it does not talk about how these metrics will be useful in determining their performance. This would not be a method that O'Neil would approve of.
Option B talks about looking at a person's familial background when designing metrics. Taking the sentence "...and discriminate against the poor and others who can't so easily game the metrics." into account, it can be inferred that O'Neil believes that a person's background could be a cause for bad performance since they don't have the same advantage that other people have. Understanding these while designing metrics and also taking them into account while judging performances would be something O'Neil would recommend.
Option C suggests taking one exceptional worker and analysing and utilizing the factors being his performance as metrics for determining others' performance. The issue with this would be that it is a very small sample size (one worker) and statistical models require a much larger sample size. Also, it is unfair to the workers from disadvantaged backgrounds to be judged based on such metrics.
Option D talks about educating workers about what metrics are used. This method is not useful in improving statistical methods and addressing their flaws. Hence, it is not something O'Neil would approve of.
Option B is the correct answer.
Choice (B)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 538

9. Refer to para 2. The narration of the incident is an anecdotal means of introducing the essence of the subject i.e. the practical applications of isotopes. The last sentence of para 2 mentions: His was the **first** use of radioactive tracers – now routine in environmental science. So choice C is correct. The other choices do not apply.
Choice (C)

10. Option A: Choice A is not true of chemical insecticides. It has been suggested in para 6 that **ionising radiation** to induce mutations in plant breeding has been used for several decades to produce new genetic lines of sorghum, garlic, wheat, bananas, beans and peppers, cereals and oil seed crops which are more resistant to pests and more adaptable to harsh climatic conditions.
Option B: This is a bane of chemical insecticides. The penultimate sentence of the fourth para states that chemical insecticides can also pose risks to non-target organisms. So, choice B is the answer.
Option C: Chemical insecticides do not catalyse the crop losses caused by insects. It has only been stated in the third sentence of para 4 that chemical insecticides have not always been effective in reducing the losses. So, choice C is not correct.
Option D: Choice D has not been suggested in the passage.
Choice (B)

11. Refer to the fifth para which talks about the uniqueness of the Sterile Insect Technique (SIT) program.
Option A: If the normal male insect fertilise the female insects in large numbers, then eradication of the insects becomes difficult. Choice A ensures that there is a greater chance of success.
Option B: The SIT program is used as a solution because insects have become resistant to chemical insecticides used. But choice B is absurd and is not the factor for success of the SIT program.
Option C: The fifth para does not talk about releasing laboratory-bred female insects or wild type male insects in infested areas. So, choice C is not the reason.

Option D: Choice D has not been mentioned. Para 5 does not talk about similarity between laboratory-bred insects and wild type male insects. It only talks about how laboratory reared male insects are rendered sterile before releasing them in the environment so that no offspring of the pest insects are produced.
Choice (A)

12. Option A: The author only mentions the positive benefits of radioisotope technology without expressing any negative features or problems about the same. So "guarded appreciation" as a response is incorrect. Choice A is not the answer.
Option B: Choice B is incorrect as the author does not express doubts about radioisotope technology.
Option C: The response of the author of the passage towards the idea of radioisotope technology can be said to be one of positive enthusiasm. Refer to the last sentence of para 1 and the first sentence of para 3. So, choice C is correct.
Option D: The passage does not criticize radioisotope technology. In fact, it states its many advantages. Hence choice D is incorrect.
Choice (C)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 540

13. The author begins by saying that we are lucky to exist. In the second paragraph, he mentions that: We're lucky life began on Earth at all, of course, and that something as complex as humans evolved. This points to choice C as the answer.
Option A: It was improbable that your parents met each other and conceived you at just the right instant, and their parents and their parents and so on back to time immemorial. But choice A runs tangent to the text and does not answer the question.
Option B: We are lucky that something as complex as humans evolved. But we cannot say that we are born because of complex biological mechanisms.
Option D: Choice D is a misdirection. The last sentence of para 1 appears twisted here. The last sentence says that we are impurities in an otherwise beautifully simple universe.
Choice (C)

14. The laws of physics themselves seem to be working against us. Ours isn't just a randomly hostile universe, it's an actively hostile universe. My physicist colleagues and I like to pretend that the laws of physics are orderly and elegant. Indeed, I just published an entire book, *The Universe in the Rearview Mirror*, about the beautiful symmetries of the universe. Programs like *Nova* or *Slate*'s own Bad Astronomy tend to focus on the knowable structure of how everything fits together.
Towards the end, the author says that it would be a mistake to be comforted by the symmetries of the universe. In truth, they are our worst enemies. Everything we know about rational, predictable arrangements dictates that we shouldn't be here at all .... The symmetry of the universe would bake us in no time at all, but an *asymmetry* rescues us.
The above sentences refute choice A and point to choice B as the answer.
Choices C and D are out of scope.
Choice (B)

15. Option A: Choice A can be inferred from the first sentence of para 3. Your existence wasn't just predicated on luck of your ancestors, but on an almost absurdly finely tuned universe.
Option B: Option B cannot be inferred. The author only mentions figuratively "Ours isn't just a randomly hostile universe, it's very hostile to our existence."
Option C: Option C can be inferred from para 3. Had the universe opted to turn up the strength of the electromagnetic force by even a small factor, poof! Suddenly stars wouldn't be able to produce any heavy elements, much less the giant wet rock we're standing on. Worse, if the universe were only minutely denser than the one we inhabit, it would have collapsed before it began.
Option D: Suddenly stars wouldn't be able to produce any heavy elements, much less the giant wet rock we're standing on. From this, we can say that the "giant wet rock" refers to the earth.
Choice (B)

16. Option A: The properties of antimatter have been discussed in the passage. For every matter, there is antimatter. The reason mentioned in the second half of choice A does not answer the question.
Option B: There is a very small difference between matter and antimatter – in terms of charge. The charge of matter is opposite to that of antimatter. But the second half of choice B is not correct.
Option C: But at the beginning of the universe, there were slightly more particles of matter rather than antimatter. We exist because "about a billion and one particles of matter were produced for every billion antiparticles". Most of the matter and antimatter annihilated each other, and so only the extra particles of matter that remained left over to form the universe. The asymmetry between matter and antimatter is why we exist in the first place. So choice C is correct.
Option D: We would not be able to tell if all matter in the universe had been replaced with antimatter this very moment. But the first sentence in choice D runs tangent to the discussion and does not answer the question.
Choice (C)

Solution for question 17:

17. The paragraph starts with how innate/ acquired distinction is commonly misunderstood as being caused by genes/ environment. Efforts to explain the distinction using arguments related to genetics have been unsuccessful and it has not been

supported by the developments in other fields like molecular developmental biology. The author then outlines three philosophical theories and explains that the fourth one belongs to sceptical tradition and questions the distinction itself. He also stresses that the distinction between innate and acquired traits might be best understood within the area of psychology only. Choice C has all the main points in the paragraph and serves as an apt summary.
Choice A is incomplete and attempts to cover only the first two sentences of the para. The second sentence of choice A is unnecessary as it covers trivial details. Choice A is also incorrect since the author does not say that the innate/acquired distinction is a psychological one. It is stated that it is best understood within the field of psychological research.
The first sentence in choice B is incorrect as the author does not say that the debate of the distinction between innate and acquired traits in an organism can be resolved. The second sentence is a bit extreme in tone. The third and fourth sentences of choice B are correct but choice B is unnecessarily wordy.
Choice D attempts to restates a particular idea in the passage but it has 'out of scope' or extraneous points. It is again incomplete as a summary.
Choice (C)

Solution for question 18:

18. The paragraph begins by saying that ammonia is repulsive to animals as it is toxic. The highlighted sentence has the contrast conjunction 'but'. We need to place this highlighted sentence ahead of a sentence which speaks positively about ammonia. The highlighted sentence can be placed only in blank (1). "actually attracted" and "Far from being repelled" in the sentence after blank (1) contrasts "repulsive" in the sentence before blank (1). The highlighted sentence forms a bridge between the sentence preceding blank (1) (It is extremely toxic) and the sentence succeeding blank (1) (Sharks are actually attracted to it). So sharks are an exception to the rule that ammonia is toxic to most marine animals and land-lubbing ones. The highlighted sentence will disrupt the thought flow if it is placed in blank 2, blank 3 and blank 4.
Choice (A)

Solution for question 19:

19. Sentence 1 borrows from another sentence with 'this reference to the Batman myth. So, this sentence obviously follows some sentence that discusses the Batman and Joker connection.
Sentence 2 talks about something that won't work – directly referencing Sentence 1 – where we are trying out something new.
Sentence 3 refers to something as 'it' – what could provide the dark genesis of the Batman story. Also, this sentence introduces the 'Batman myth', which means 31 is a block.
Sentence 4 is an independent sentence about 'Joker's popularity'. 4 provides the context for 3 – Joker's popularity provides the dark genesis of the Batman story. Similarly, 2 follows 1, and 2 talks about why it wouldn't work. 1 is directly dependent on 3. So, **4312.**
Ans: (4312)

Solution for question 20:

20. Option A: This option equates reality with contemporary reality, which is fine, but what fiction does really well – give us experiential benefits – is not 'alternative reality'. The option equates alternative experience to understand things better to alternative reality. Option A is not the answer.
Option B: This option gives both the ideas – that there are 'some' (not all) art forms which do a better job than fiction in depicting contemporary ideas and reality, and that fiction helps people explore and experience to understand things better. Option B is the answer.
Option C: Fiction helps one experience - true. But not experience 'sensations they wouldn't otherwise explore'. Such a comparison has not been made in the para, that fiction opens doors they wouldn't otherwise approach. Option C is not the answer.
Option D: While this is partially true, it is not the entire idea. It is just taking one small part of the para – whose essence is that fiction helps one explore even if it doesn't represent contemporary reality well. Option D is, therefore, close but not the answer.
Choice (B)

Solution for question 21:

21. On a careful reading of the sentences, it can be observed that sentence 4 is a general sentence that begins the paragraph. It introduces the background: Former champions were missing from this year's US Open and less-familiar names wished to make a name for themselves. Sentence 4 is followed by sentence 3. From "a throng of less-familiar names" (sentence 4), "one name is particularly surprising: Diego Schwartzman" (sentence 3). Sentence 3 is followed by sentence 2. "Diego Schwartzman who is just five feet and seven inches tall" in sentence 3 links with "Mr Schwartzman, nicknamed "El Peque" ("the small one")" in sentence 2. Sentences 2 and 1 form a mandatory pair. "once before reached the third round of a major" in sentence 2 links with "he had won fewer than half of his matches on hard courts". Sentence 1 concludes the para. Hence, 4321.
Ans: (4321)

Solution for question 22:

22. The paragraph begins by discussing the history behind the idea of gravitational waves.

We need to place the highlighted sentence ahead of a sentence which speaks about approximations.

The highlighted sentence can be placed only in blank (2). "It is a simple idea, but the equations that give it mathematical heft are damnably hard to solve." in the sentence prior to blank 2 links with "Only by making **certain approximations** can solutions be found." as given in the highlighted sentence. So, the highlighted sentence forms a bridge between the sentence preceding blank (2) (It is a simple idea, but the equations that give it mathematical heft are damnably hard to solve.) and the sentence succeeding blank (2) (And **one such approximation** led Einstein to an odd prediction: any accelerating mass should make ripples in spacetime. Hence (2) is the answer.

The highlighted sentence will disrupt the thought flow if it is placed in blank 1. The sentence "Mass, Einstein realized, deforms the space and time around itself." before blank (1) need to be immediately followed by the sentence: "Gravity is the effect of this, the behaviour of objects dutifully moving along the curves of mass-warped space-time". Hence (1) is not the answer.

The highlighted sentence will be redundant in blanks (3) and (4). The second part of the paragraph mainly talks about the thoughts and actions of Einstein and all the given sentences in the second para need to run continuously.
Choice (B)

Solution for question 23:

23. Sentence 1 starts with a cause-effect indicator 'As though', which means it is following up from the previous sentence.
It is also an introspective thought about what one sees looking inwards.
Sentence 2 talks about 'these tiny keyholes' connecting to a sentence that has introduced the keyholes.
Sentence 3 talks about what 'you' already knows – the difference between size and speed of everything that flashes through you.
Sentence 4 talks about a particular truth, about what something is like. A sentence of this nature has to precede other ideas since it has a tone of anticipation, not a tone of conclusiveness. You wouldn't say 'you already know what it's like' after you have explained what it's like.
Sentence 1 introduces/explains the idea of keyholes which means 12 is a block, since 2 has a reference to these keyholes. Sentence 1 starts with 'as though' – carrying forward an explanation or rather the continuation of an explanation.
Sentence 3 is the bridge between 4 and 1. 4 starts with a vague reference to truth, and only 3 can follow it because it repeats the same sequence of words 'you already know.'
Ans: (4312)

Solution for question 24:

24. Option A: This is true as a leader, according to the para should be cognizant of the importance of diversity when it comes to ideating and planning especially during solving complex problems while also ensuring that convergence or unity of perspective is important when things have to be done quite fast and efficiently. A leader understands when diversity helps and when it becomes an impediment. Option A is the answer.
Option B: A good leader understands when to encourage diversity, true. However, the question of 'discouraging diversity' has not really been discussed. The para was more inclined towards giving importance to convergence (which is not necessarily an act of discouraging diversity). Hence, Option B is not the answer.
Option C: The para talks more about the phase when diversity must be encouraged and the phase when it will take a back seat. This option seems to suggest two different scenarios, as to when diversity is an advantage and when it is a disadvantage. It is not that diversity is useful only when the problem is complex. That is merely an example. And it is not that diversity is a disadvantage only if there is a looming deadline. So the option looks at diversity as an entirely different concept compared to the para.
Option D: While the para does talk about 'managing' diverse perspectives, it doesn't really talk about focusing on a 'unified goal'. The author mentions specific scenarios – looming deadlines – during which a leader should push the team towards 'convergence' or agreement rather than a 'goal' per se. Hence, Option D is not the answer.
Choice (A)

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	22
Medium	1, 2, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 18, 19, 20
Difficult	3, 5, 21, 23, 24
Very Difficult	12, 17

SECTION – II

Solutions for questions 1 to 5:

From the given information

	Maths	Physics	Chemistry
Abhijit	✓	×	
Dheeraj	✓		×
Mohit	×		
Total	2	1	1

(As the total number of students who passed in Maths is two, Abhijit must have definitely passed in Maths.)

1. It is given that both the students who failed in Physics also failed in Chemistry, hence the student who passed in Physics must also have passed in Chemistry. That student must be Mohit.

Hence the final table is as follows.

	Maths	Physics	Chemistry
Abhijit	✓	×	×
Dheeraj	✓	×	×
Mohit	×	✓	✓

From the above table, only statement (D) is true.
Choice (D)

2. As Mohit passed in Physics, we get the following table.

	Maths	Physics	Chemistry
Abhijit	✓	×	
Dheeraj	✓	×	×
Mohit	×	✓	

From the above table, only (A) could be true.
Choice (A)

3. From the given information,
(i) Exactly one of Dheeraj and Mohit passed in Physics, hence (A) cannot be true.
(ii) Mohit failed in Maths, hence (B) cannot be true.
(iii) Exactly one of Mohit and Abhijit must have passed in Chemistry, hence (C) cannot be true.
(iv) Dheeraj could have passed in only Maths and Mohit in Physics and Chemistry. (D) can be true.
Choice (D)

4. Given that Mohit passed in exactly two subjects

	Maths	Physics	Chemistry
Abhijit	✓	×	×
Dheeraj	✓	×	×
Mohit	×	✓	✓

From the above table, choice (C) must be true.
Choice (C)

5. The persons who passed in the same subjects must be Abhijit and Dheeraj. Both of them must have passed in Maths but not in any other subject.

	Maths	Physics	Chemistry
Abhijit	✓	×	×
Dheeraj	✓	×	×
Mohit	×	✓	✓

From the above table, choice (D) must be true.
Choice (D)

Solutions for questions 6 to 10:

Among the four persons who was serviced by a hairdresser, the person who spent the least time would have been serviced first.
Since John takes between 11 minutes to 15 minutes to cut one person's hair, Akhil must have been the first to be serviced by John.
Similarly, Hardip must have been serviced by Jack. Either Narendra or Govind would have been serviced by Gary.

Since Akhil was serviced by John, the next person to have been serviced by John would have spent at least 13 + 11 = 24 minutes and at most 13 + 15 = 28 minutes in the saloon.

From the table, Narendra must be the second person to have been serviced by John.

Hence, Govind is the first person that Gary serviced.

The second person that Jack would have serviced waited for at least 17 + 16 = 33 minutes and at most 17 + 20 = 37 minutes. Either Ram or Das can be this person.

The third person that John serviced would have spent at least 24 + 11 = 35 minutes and at most 24 + 15 = 39 minutes. From the table, we can see that only Das can be this person. The fourth person to be serviced by John must have spent at least 47 minutes and at most 51 minutes. Neeraj has to be the fourth person.

The second person serviced by Gary must have spent at least 43 minutes and at most 47 minutes. From the table, Jayant is the second person serviced by Gary.

The third person serviced by Gary must have spent at least 64 minutes and at most 67 minutes. Gautam must be the third person to be serviced by Gary. The last person to be serviced by Gary must have spent at least 88 minutes and at most 92 minutes. Shiv has to be the fourth person serviced by Gary.

Since Das is serviced by John, Ram must be the second person to have been serviced by Jack. The third person serviced by Jack must be Anant and the fourth person to be serviced by Jack must be Nitin.

The following table provides the order in which each of the three barbers serviced the customers and the time taken by them for servicing each customer:

John	
Person	Service Time
Akhil	13
Narendra	11
Das	12
Neeraj	14

Jack	
Person	Service Time
Hardip	17
Ram	16
Anant	19
Nitin	17

Gary	
Person	Service Time
Govind	22
Jayant	21
Gautam	24
Shiv	23

6. Anant was serviced by Jack. Choice (B)

7. John took a maximum of 14 minutes to service any customer. Ans: (14)

8. Ram was not serviced by John. Choice (D)

9. Among the given options, Neeraj waited for the maximum amount of time (36 minutes).
Choice (A)

10. The average time to service a customer by Gary = 90/4 = 22.5 minutes. Ans: (22.5)

Solutions for questions 11 to 15:

11. The population decreased by more than 20% in 1953, 1969, 1976 and 1978, i.e., for 4 years.
Ans: (4)

12. By observing the graph, we can see that the percentage increase was the highest in 1982 (100%).
Choice (B)

13. Number of years in which the population was 10 = 4
Number of years in which the population was 15 = 7
Number of years in which the population was 20 = 10
Number of years in which the population was 25 = 9
Number of years in which the population was 30 = 3
Number of years in which the population was 35 = 8 (including 1990)
Average
$= \frac{4\times10+7\times15+10\times20+9\times25+3\times30+8\times35}{41} = 22.93$
Choice (B)

14. The population of the tortoises was greater than 20 from 1958-72 and 1986-90.
Therefore, the population of the tortoises was greater than 20 for 15 + 5 = 20 years
$x = 20$
Required percentage $= \frac{20}{41} = 48.78\%$
Choice (A)

15. In 1950, there were 20 tortoises. However, 1978, the number of tortoises reduced to 10. Hence, only 10 tortoises would have survived for the entire duration.
Ans: (10)

Solutions for questions 16 to 20:

From (ii), only one student (D) had a change in rank of 3.
Also, from (vi), the revised ranks of all students are unique.
Assume D's initial rank is n. Then D's revised rank must be n + 3.
If the next initial rank is n + 1, then the revised rank of that student will be n + 4, which is not possible as this student also would have a change in rank of 3.
Hence, the next initial rank must be n and the next revised rank must be n + 4.
The student with revised rank of n + 2 cannot have an initial rank of n – 1 since the difference will be 3. Hence, this student's initial rank must also be n.

Hence, the initial and revised ranks of D and the students immediately above and below him (as per the revised ranks) will be as follows:

Student	Initial Rank	Revised Rank
	n	n + 2
D	n	n + 3
	n	n + 4

The value of n cannot be 1 (since there must be more than three students with rank of 1, for n to be 1).
If n = 2, then the three students will have revised ranks of 4, 5 and 6.
For a student to have an initial rank of 2 and revised rank of 4, there must be three students with initial rank of 1. These three students will have revised ranks of 1, 2 and 3.
The students with revised ranks of 7 and 8 can have initial ranks of (3, 3) or (3, 4). If the initial ranks of these two students are (3, 4), then two students will have maximum change in rank, which violates condition (i).
Hence, the last two students should have initial ranks of (3, 3).
This is one possible case.
If the value of n is 3, then D's revised rank must be 6.
This is not possible as A's revised rank is 6.
If the value of n is 4, then the three students in the table above will have revised ranks of 6, 7 and 8.
D's revised rank will be 7 and G's revised rank must be 8 (for the change in rank to be maximum for G).
Also, A's revised rank must be 6 (from (iii)). C's revised rank should be less than that of A. But this is not possible in this case. Hence, the value of n cannot be 4.
Hence, the only possible value of n is 2 and as per the above explanation, the initial and revised ranks of the eight students are as follows:

Initial Rank	Revised Rank
1	1
1	2
1	3
2	4
2	5
2	6
3	7
3	8

We know that D's revised rank is 5, A's revised rank is 6 and G's revised rank is 8 (because the change in rank is the maximum).
C's revised rank must be 7 (since C's revised rank is worse than that of A).
B's revised rank must be 2. The revised rank of F and H can be 1 and 3, in any order.

Initial ranks:

1	H/F	B	F/H
2	E	D	A
3	C	G	

Revised ranks:

1	F/H
2	B
3	H/F
4	E
5	D
6	A
7	C
8	G

16. Revised rank of D is 5. Ans: (5)

17. B, H and F got 1st rank initially. Choice (C)

18. C's rank changed by four. Ans: (4)

19. Either F or H would have got a revised rank of 1.
Choice (D)

20. Among the persons who got the same rank, the person with least attempts got a better rank.
∴Attempts of F/H < B < H/F
E < D < A
C < G
∴ D cannot be the person with least attempts as it is certain that he attempted more questions than E.
Choice (C)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	1, 2, 3, 4, 5, 11, 12, 13, 14
Medium	6, 7, 8, 9, 10, 15
Difficult	16, 17, 18, 19, 20
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1.

	A		B
Efficiency ratio	1	:	4
Time ratio	4	:	1
Time taken	4t		t (Difference 3t = 15)

Therefore, A takes 20 days and B takes 5 days to complete the work.
They will together complete the work in $\frac{1}{\frac{1}{20}+\frac{1}{5}}$
= 4 days.
Choice (A)

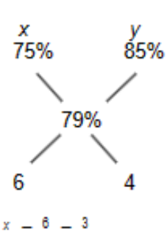
2. $7^n = (6+1)^n = 1 + 6n + {}^nC_2(36) + 216[{}^nC_3 + {}^nC_4(6)$ + -------+ ${}^nC_n(6)^{n-3}]$
$= 1 + 6n + 36\frac{n(n-1)}{2} + 216[{}^nC_3 + {}^nC_4(6)$ ----- + ${}^nC_n(6)n - 3]$
$\therefore 7^n - 6n(3n-2) - 7$
$1 + 6n + 18n^2 - 18n + 216(K) - 18n^2 + 12n - 7$
$= 216(K) - 6$
Now, when 216(K) – 6 is divided by 216 the remainder is 210.

Alternative Solution:

Though the question says n > 5, one could figure out (or guess) that the condition that n > 5 is not significant. Hence, by trying n = 0, 1, 2, one can figure out that option (C) fits.
Choice (C)

3. Let the quantities of the two detergents A and B mixed by x and y respectively

Using Alligation, we get



$\therefore \frac{x}{y} = \frac{6}{4} = \frac{3}{2}$
Choice (D)

4. It is given that, p, p + 2q, 3p + q and 30 are in Arithmetic Progression.
$\therefore p + (3p + q) = 2(p + 2q)$
$2p = 3q$
Therefore, the terms (in terms of q) are $\frac{3q}{2}, \frac{7q}{2}, \frac{11q}{2}$ and $\frac{15q}{2}$
But $\frac{15q}{2}$, i.e., the fourth term = 30 (given)
⇒ q = 4 and p = 6, i.e., 1st term = 6
and common difference = 2p = 8.
Thus the 2016th term = 6 + (2016 – 1)8 = 16126
Ans: (16126)

5. Let the cost price of B be x
⇒ The cost price of A is 0.85x
Thus, the selling prices of A and B are 1.15 × 0.85x and 1.12 × x.
Required ratio
$= \frac{1.15\times0.85x}{1.12\times x} = \frac{0.9775}{1.12} = \frac{9775}{11200} = \frac{391}{448}$
Choice (C)

6. Let the speeds of trains be x m/sec and y m /sec, where x > y.
$\frac{500+300}{x-y} = 80$
⇒ x – y = 10 __ (1)
and $\frac{500+300}{x+y} = 20$
⇒ x + y = 40 ___ (2)
Solving (1) and (2),
x = 25 m/sec and y = 15 m/sec
∴ Speed of slower train $= 15\times\frac{18}{5}$ = 54 km/hr
Ans: (54)

7. The compound interest for the (n + 1)th year is R% more than that for the nth year, where R% is the annual rate of interest. Since the interest for the 5th year is 10% more than that of the 4th year, it is clear that R = 10
The compound interest for the 3rd year $= \frac{2420}{1.1}$
i.e., ₹2200
So, the compound interest for the 2nd year $= \frac{2200}{1.1}$
= ₹2000
Choice (B)

8. (k – 1).(k – 2).(k – 3)...(3).(2).(1) = (k – 1)!
(n – 1)! is not divisible by n if n is prime. There are 10 prime numbers in the given range. In addition to above 10 numbers, the given condition is also satisfied for n = 4,.i.e., a total of 11 numbers.
Ans: (11)

9. $((2 * 3) * 3) = 2^3 * 3 = 3^8 = 6561$
$((3 * 2) * 2) = 2^3 * 2 = 2^8 = 256$
∴ required sum = 6561 + 256 = 6817.
Ans: (6817)

10. Net Speed of the boat upstream = v
Net Speed of the boat downstream = $4v$
Time taken Upstream $= \frac{d}{v}$
Time take Downstream $= \frac{d}{4v}$
Total time $= \frac{d}{v} + \frac{d}{4v} = 2\times60$ minutes (given)
$\Rightarrow \frac{5d}{4v} = 120 \Rightarrow$ difference of time $= \frac{3d}{4v} = \left(\frac{5d}{4v}\right)\times\frac{3}{5}$
$= 120\times\frac{3}{5} = 72$
Therefore, the boat took 72 minutes more while travelling upstream than what it took to travel downstream.
Choice (D)

11. Since $0 < x < \frac{1}{2}$, let $x = \frac{1}{4}$.
From the options, we get
(A) $x^{-2} = \frac{1}{x^2} = \frac{1}{\left(\frac{1}{4}\right)^2} = 16$
(B) $x^{-\frac{3}{2}} = \frac{1}{x^{\frac{3}{2}}} = \frac{1}{\left(\frac{1}{4}\right)^{\frac{3}{2}}} = 4^{\frac{3}{2}} = 8$
(C) $\frac{1}{\sqrt{x}} = \frac{1}{\sqrt{\frac{1}{4}}} = \pm 2$ (assuming positive root as per convention)
(D) $\left(\frac{1}{x^{-2}}\right)^{-3} = (x^2)^{-3} = x^{-6} = \frac{1}{x^6} = \frac{1}{\left(\frac{1}{4}\right)^6} = 4096$
Therefore, choice (D) gives the greatest value.

Alternative Solution:

Let $y = \frac{1}{x}$, where y > 2
Considering the options, we are required to compare
y^2, $y^{\frac{3}{2}}$, $y^{\frac{1}{2}}$ and y^6. Since y > 2, clearly y^6 will be the greatest.
Choice (D)

12. If a positive number a is expressed as the sum of two positive numbers t_1 and t_2 then $\lfloor a \rfloor$ could be at most be two less than $\lceil s_1 \rceil + s_2$, i.e., the fractional parts of $\lceil s_1 \rceil - s_1$, $\lceil s_2 \rceil - s_2$ and $a - \lceil a \rceil$ together can provide two.
Similarly, the fractional parts of $\lceil s_1 \rceil - s_1$, $\lceil s_2 \rceil - s_2$, ..., $\lceil s_{11} \rceil - s_{11}$ and $a - \lceil a \rceil$ together can provide eleven.
∴ The minimum value of m is 11.
Similarly, the least value of n is 7.
∴ $(m + n)_{min} = 11 + 7 = 18$.
Choice (D)

13. Let the cost price of the 236 pens be P
⇒ Selling price of 200 pens = P
Selling price of the remaining 36 pens $= 36\frac{P}{200}$
Total selling price of the 236 pens $= P + \frac{36P}{200}$
Profit = SP – CP $= P + \frac{36P}{200} - P = \frac{36P}{200}$
Profit% $= \frac{\left(\frac{36P}{200}\right)}{P}\times100 = 18\%$.
Choice (A)

14. It is given that, $\sqrt{x} \ge 3x$
Let $x = t^2$, so $t > 3t^2$
$\Rightarrow 3t^2 - t < 0$
$\Rightarrow t(3t - 1) < 0$

Therefore $0 \le t \le \frac{1}{3}$ satisfies the above inequality.
The maximum value of t is $\frac{1}{3}$ and the maximum value of x is $\frac{1}{9}$ ($\because x = t^2$)
Choice (C)

15. (3m + 4w) × 4 = (4m + 2w) × 6
⇒ 12m + 16w = 24m + 12w
⇒ 4w : 12m
⇒ w = 3m
Thus one woman-day is equal to three man-days
3m + 4w ≡ 15 m
Thus, the work requires 15 × 4 = 60 man-days
9m + 2w = 15m
Required days $= \frac{60}{15} = 4$
Ans: (4)

16. Discriminant $= (2b)^2 - 4(a)(c) = 4(b^2 - ac) = 0$
since the roots are equal.
$\therefore b^2 = ac$
a < b < c with a, b and c being single digit natural numbers.
∴ a = 1 or 2 or 4.
Using statement I, a = 1 or 4. I is not sufficient.
Using statement II, a = 1 or 2. II is not sufficient.
Using both I and II, a = 1
Ans: (4)

17. The number N has 28 factors.
28 {or $2^2(7)$} itself has 6 factors. The different ways in which 28 can be expressed as the product of 1, 2 or 3 factors, the corresponding structures for N, and the number of factors of N which are perfect squares, are tabulated below:

Different ways of writing 28	Structure of N	Factors which are perfect squares	No. of factor that are perfect squares
1(28)	p^{27}	$p^0, p^2, p^4.......p^{26}$	14
2(14)	p^1q^{13}	$p^0q^0, p^0q^2, ...p^0q^{12}$	(1)(7) = 7
4(7)	p^3q^6	$p^0q^0 ... p^0q^6, p^2q^0... p^2q^6$	(2)(4) = 8
2(2)(7)	$p^1q^1r^6$	$p^0q^0r^0, p^0q^0r^2, ..., p^0q^0r^6$	(1)(1)(4)=4

∴ N can have 4, 7, 8 or 14 factors which are perfect squares but not 6 such factors.
Choice (B)

18. The sum of all 100 numbers = 60 × 40 + 40 × 60 = 4800
Therefore, the average of all 100 numbers $= \frac{4800}{100} = 48$
Choice (A)

19. Let the first term of the GP be x and the common ratio be y.
Given $p = x.y^{a-1}$
$q = x.y^{b-1}$
$r = x.y^{c-1}$
$\Rightarrow p^{b-c}.q^{c-a}.r^{a-b}$
$= [x^{b-c}.(y^{a-1})^{b-c}][x^{c-a}.(y^{b-1})^{c-a}][x^{a-b}.(y^{c-1})^{a-b}]$
$= x^{b-c+c-a+a-b}.y^{(ab-ac-b+c+bc-ab-c+a+ac-bc-a+b)}$
$= x^0.y^0 = 1$

Alternative Solution:

Assuming a simple G.P. as 1, 1, 1 and 1, 2, 4, it can be observed from the choices that only choice (B) is possible.
Choice (B)

20. There are three possibilities. These are explained below.
(i) Possibility 1: The numbers are of the form 24xy.
24xy = 24(100) + xy
24(100) is divisible by 24.
∴xy must be divisible by 24.
∴xy = 00 or 24 or 48 or 72 or 96.
∴There are 5 possible numbers.
(ii) Possibility 2: The numbers are of the form x24y.
x24y must be divisible by both 8 and 3. As it is divisible by 8, 24y is divisible by 8. ∴y = 0 or 8.
x24y has the sum of its digits as x + y + 6. This must be divisible by 3.
∴ x + y must be divisible by 3. When y = 0, x must be divisible by 3. ∴x = 3 or 6 or 9.
When y = 8, x + 8 must be divisible by 3.
∴x = 1 or 4 or 7.
A total of 6 numbers are possible.
(iii) Possibility 3: The numbers are of the form xy24.
xy24 = 100(xy) + 24
100(xy) must be divisible by 24.
100 is divisible by 4.
∴ xy must be divisible by 6.
∴ xy has 15 possibilities (xy = 12 or 18 or ...... 96)
∴ A total of 15 numbers are possible.
But 2424 is occurred in possibility (i) as well as possibility (iii).
∴ A total of 5 + 6 + 15 – 1 = 25 numbers are possible.
Ans: (25)

21. Let the cost of a pen be 24k
∴the cost of an eraser = 2k
∴the cost of a pencil = 3k
It is given that, 2(24k) – 5(3k) = 33k = 33
∴k = 1
∴The cost of one pen, two pencils and three erasers
= 24k + 2(3k) + 3(2k) = 36k = ₹36
Choice (B)

22. N = Largest possible number
= H.C.F. (171 – 3, 247 – 7)
= H.C.F. (168, 240) = 24
When 24 divides 800, the remainder is 8.
Ans: (8)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	9, 11, 18
Easy	1, 3, 4, 5, 6, 7, 8, 13, 14, 15, 21, 22
Medium	2, 10, 12, 16, 19
Difficult	20
Very Difficult	17

Done
Tip: Scroll slowly for the best results